

11.24

a. Mixed is significant at 5% level with

$p_value = 0.0032 < 0.05$

Joint significance of Stormy and Mixed:

$F = 16.14 > 10$, $p_value < 0.05$

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
mon          -0.10825    0.10339  -1.047  0.29753
tue          -0.06609    0.10103  -0.654  0.51446
wed          -0.04927    0.10377  -0.475  0.63591
thu           0.03935    0.10071   0.391  0.69681
stormy       0.44634    0.07921   5.635 1.51e-07 ***
mixed       0.23688    0.07851   3.017  0.00321 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3413 on 104 degrees of freedom
Multiple R-squared:  0.245,    Adjusted R-squared:  0.2014
F-statistic: 5.624 on 6 and 104 DF,  p-value: 4.349e-05
```

```
> #join test
> linearHypothesis(reduced,c("stormy=0","mixed=0"))
```

Linear hypothesis test:

```
stormy = 0
mixed = 0
```

Model 1: restricted model

Model 2: $lprice \sim mon + tue + wed + thu + stormy + mixed$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b. The $lprice$ coefficient now (-0.93014) is lower than in table 11.5 (-1.1194).

However, the coefficient is still significantly negative.

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533 < 2e-16 ***
lprice      -0.93014    0.35340  -2.632  0.00977 **
mon          -0.01190    0.20837  -0.057  0.95456
tue          -0.52583    0.20230  -2.599  0.01068 *
wed          -0.56262    0.20696  -2.719  0.00767 **
thu           0.09987    0.20285   0.492  0.62350
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

c. Test the validity of the surplus instrument:

H_0 : instruments are valid

Sargan: $p_value = 0.380 > 0.05 \rightarrow$ do not reject $H_0 \rightarrow$ IVs are valid.

```
Diagnostic tests:
              df1 df2 statistic  p-value
Weak instruments  2 104   16.140 7.86e-07 ***
Wu-Hausman       1 104    1.489   0.225
Sargan           1  NA    0.772   0.380
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d. Joint test for mon, tue, wed, thu:

$F = 0.67 < 10$; $p_value = 0.64 > 0.05$

$\rightarrow$ We fail to reject the null hypothesis.

Therefore, the variables mon, tue, wed and thu are not jointly significant in the reduced-form equation for $\ln(PRICE)$.

$\rightarrow$ The variables mon, tue, wed and thu do not explain price well in reduced-form.

$\rightarrow$ Could not confident to estimate the supply equation by 2SLS.

Linear hypothesis test:

```
mon = 0
tue = 0
wed = 0
thu = 0
```

Model 1: restricted model

Model 2: $lprice \sim mon + tue + wed + thu + stormy + mixed$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

11.28

a.

- Supply: $Q = \beta_1 + \beta_2 P + \beta_3 PF + e_s \rightarrow P = b_1 + b_2 Q + b_3 PF + u_s$

- Demand: $Q = \alpha_1 + \alpha_2 P + \alpha_3 PS + \alpha_4 DI + e_d \rightarrow P = a_1 + a_2 Q + a_3 PS + a_4 DI + u_d$

b. All the estimated coefficients are different from zero.

_Supply: $P = b_1 + b_2 Q + b_3 PF + u_s$

Q is positive $\rightarrow$ P increases when Q increases (law of supply)

PF is positive $\rightarrow$ P increases when PF increases. (cost increase $\rightarrow$ price increase)

```
Call:
lmvreg(formula = p ~ q + pf | ps + di + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-9.7983 -2.3440 -0.6281  2.4350 11.1600

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -58.7982     5.8592  -10.04 1.32e-10 ***
q             2.9367     0.2158   13.61 1.32e-13 ***
pf            2.9585     0.1560   18.97 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared:  0.9486,    Adjusted R-squared:  0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16
```

_Demand: $P = a_1 + a_2 Q + a_3 PS + a_4 DI + u_d$

Q is negative $\rightarrow$ P increases when Q decreases (law of demand)

PS is positive $\rightarrow$ P increases when PS increases (PS expensive $\rightarrow$ people buy more truffles $\rightarrow$ P increase)

DI is positive $\rightarrow$ P increases when DI increases (inflation)

```
lmvreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-39.661  -6.781   2.410   8.320  20.251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.428     13.592  -0.841  0.40810
q             -2.671     1.175  -2.273  0.03154 *
ps             3.461     1.116   3.103  0.00458 **
di            13.390     2.747   4.875  4.68e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared:  0.5567,    Adjusted R-squared:  0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06
```

c.

$a_2 = -2.671 \Delta P / \Delta Q \rightarrow \Delta Q / \Delta P = -1/2.671$

Elasticity = $\Delta Q / \Delta P * p_{\text{mean}} / q_{\text{mean}} = 1.27$

```
> a2 <- coef(dem_iv)
>
> p_mean <- mean(truffles$p)
> p_mean
[1] 62.724
> q_mean <- mean(truffles$q)
> q_mean
[1] 18.45833
> E <- (1/a2["q"])*(p_mean/q_mean)
> E
      q
-1.272464
```

d.

$DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$

_Supply: $P = -58.7982 + 2.9367Q + 2.9585*23 = 9.2473 + 2.9367Q$

_Demand: $P = -11.428 - 2.671Q + 3.461*22 + 13.39*3.5 = 111.579 - 2.671Q$

```
> supply_intercept
(Intercept)
  9.246963
> supply_slope
      q
2.936711
> demand_slope
      q
-2.670519
> demand_intercept
(Intercept)
 111.5801
```

e. Equilibrium: $9.2473 + 2.9367Q = 111.579 - 2.671Q$

```
> e_q <-(demand_intercept-supply_intercept)/(supply_slope-demand_slope)
> e_p <- demand_intercept + demand_slope*e_q
> e_q
(Intercept)
18.25021
> e_p
(Intercept)
62.84257
```

Predict:

Q = 18.2604

P = 62.8154

- ➔ Almost equal to the equilibrium values.
- ➔ They agree very closely.

```
> #predict
> rf_q <-lm(q ~ ps + di + pf, data = truffles)
> rf_p <-lm(p ~ ps + di + pf, data = truffles)
> pred <- data.frame(ps=PS0, di=DI0, pf=PF0)
> pred_q <- predict(rf_q,newdata = pred )
> pred_p <- predict(rf_p,newdata = pred )
> pred_q
1
18.2604
> pred_p
1
62.81537
~ |
```

f.

Supply:

```
Call:
lm(formula = p ~ q + pf, data = truffles)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.4721	-3.3287	0.1861	2.0785	10.7513

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-52.8763	5.0238	-10.53	4.68e-11 ***
q	2.6613	0.1712	15.54	5.42e-15 ***
pf	2.9217	0.1482	19.71	< 2e-16 ***

Demand:

```
lm(formula = p ~ q + ps + di, data = truffles)
```

Residuals:

Min	1Q	Median	3Q	Max
-25.0753	-2.7742	-0.4097	4.7079	17.4979

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-13.6195	9.0872	-1.499	0.1460
q	0.1512	0.4988	0.303	0.7642
ps	1.3607	0.5940	2.291	0.0303 *
di	12.3582	1.8254	6.770	3.48e-07 ***

For demand, the Q coefficient should be negative (law of demand) -> wrong in terms of logic.

For supply, the sign is right but P and Q are simultaneity so should use OLS, instead could use 2SLS.